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Midterm Activity #1
Systematic Literature Review
Planning

In Partial Fulfillment of the
Requirements for the Course
CS324: Methods of Research

EA Technologies

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PICOC Protocol Definition

Population

Philippine government employees and the existing government email infrastructure they rely on, characterized by a heterogeneous and resource-constrained IT environment. Based on observations about the real-world conditions in the Philippine government. The population is not a uniform group of users with modern, company-managed devices. It is a complex ecosystem with several distinct layers:

1. **End-Users (The Employees):**
    - **The "Legacy" User:** Employees stuck using old, unpatchable desktops (e.g., Windows 7, or even Windows XP) that cannot run modern security software or browser extensions. The Philippine Institute for Development Studies has documented that "inadequacy of ICT infrastructure" and "lack of hardware" remain persistent barriers to e-government implementation (Serfino et al., 2020), while the DICT itself acknowledges that government faces "longstanding productivity barriers such as legacy systems" (DICT, 2023). Compounding this, Commission on Audit reports reveal that billions in ICT budgets go unutilized—the Office of the President alone left 72.48% (P233.1 million) of its 2024 ICT budget unspent, preventing necessary upgrades (COA, 2024).
    - **The "BYOD" User:** Employees who must use their personal mobile phones (Android/iPhone) to check email because they lack a government-issued device or are working in the field. You cannot install corporate security tools on these personal devices. Legally speaking in the Philippines, BYOD is "a workplace setup where employees use personally owned smartphones for work" (Respicio & Co., 2024). The National Privacy Commission's own guidelines acknowledges this reality, stating that personal devices "may be used if provision of organization-owned ICT resources is impractical" but such practice "must be governed by the organization's BYOD policy" (NPC, 2021). Legal experts note that requiring device management software on personal phones creates "questionable" consent due to employer-employee power imbalances under the Data Privacy Act (Respicio & Co., 2024).

- **The "Cross-Platform" User:** A mix of employees on Macs, Linux machines, or tablets, all with different operating systems and email clients, making a uniform, device-based security solution nearly impossible. The DICT documents that government operations have historically been "siloed" with no "centralized systems" (DICT, 2023), and PIDS research identifies "lack of interoperability or lack of shared standards and compatible infrastructure across government agencies" as a major barrier (Serfino et al., 2020). The scale of this heterogeneity is evident in DICT's Project SONAR, which discovered over 30,000 vulnerabilities across at least 2,002 government digital assets with widely varying configurations (DICT, 2024).

2. **Technical Environment:**
    - **The Existing Email Servers:** These are the legitimate, government-owned email servers (e.g., at DICT, DEPED, DOST) that currently receive mail from the internet and deliver it directly to the inboxes of the users described above. The GovMail system, mandated by CSC-NCC Resolution 2010-01, provides official "dot.gov.ph" email addresses to government employees (CSC-NCC, 2010). However, a 2024 security analysis revealed that 78% of Philippine government domains lack SPF records, 87.52% have no DMARC record at all, and less than 10% have correct DMARC configuration. Meaning these servers deliver mail without basic authentication protocols (PowerDMARC, 2024).
    - **The Network Architecture:** The existing pathway for email, which currently has no security checkpoint before the inbox. The Philippines experienced the highest number of cyberattacks in Southeast Asia in 2023, with 75% of organizations affected (PowerDMARC, 2024). DICT Undersecretary David Almirol testified that his agency identified 20,144 vulnerabilities in government networks as of August 2024 (Almirol, 2024), leading to the adoption of the National Cybersecurity Plan 2023-2028 specifically to address "the rising number of cyber threats in the Philippines" (Marcos, 2023).

3. **Organizational Context:**
    - **Limited IT Budget:** The government cannot afford to purchase, deploy, and maintain endpoint security licenses for potentially hundreds of thousands of devices. The cybersecurity workforce crisis illustrates this: DICT Secretary Ivan Uy revealed that private sector cybersecurity experts earn P200,000 monthly while government experts earn only P50,000. Which is a 75% pay gap that makes talent retention impossible (Uy, 2024). During Senate hearings, it was revealed that even critical agencies like the Anti-Money Laundering Council had their proposed P125.56 million cybersecurity budget cut to P93.01 million, with Senator Lacson noting these amounts are "smaller than the commissions demanded by corrupt officials" in other agencies (Lacson, 2021). The DICT itself admits it struggles "due to lack of funding and personnel" to mitigate attacks (Uy, 2024).
    - **Lack of Centralized Device Control:** IT admins often do not have administrative rights or the ability to enforce security configurations on every device, especially personal ones. The Data Privacy Act (R.A. 10173) creates legal barriers—employer "consent" for device monitoring is "legally and practically questionable because of the power imbalance" (Respicio & Co., 2024). Legal experts distinguish between "light BYOD" (authenticator apps, defensible) and "deep BYOD" (MDM with broad permissions, higher legal risk), meaning government cannot legally enforce the level of control needed for endpoint security on personal devices (Respicio & Co., 2024). Combined with the technical heterogeneity documented by PIDS and DICT, uniform device-level control is both legally prohibited and technically impossible.

Intervention

A centralized, server-side AI-powered email gateway that is inserted into the email flow to intercept, analyze, and act upon all incoming messages before they reach the end-user's device or the government's internal email server.

1. **Location:**
   - A physical and logical positioned barrier between the public internet and the government's mail server. Acting as a mandatory checkpoint for all emails, without exception, must pass through in order to reach its destination. This architectural approach aligns with what industry refers to as a "recipient-centric gateway sitting at the corporate network perimeter" (Hartman, 2016). The strategic positioning creates what Microsoft documentation describes as a centralized mail system where "all Exchange servers are placed in a central data center for management," offering the most cost-effective and easily manageable model (Microsoft, 2005). This centralized checkpoint ensures that security is applied uniformly before any email reaches its final destination.

2. **Core Functions:**
   - **Interception:** The gateway receives email traffic first, acting as the initial point of contact for the government's mail domain. This follows the established pattern of email concentrators that "receive email associated with multiple email accounts from multiple email servers" before any distribution occurs (Campbell & Solyanik, 2002). Modern implementations use "ingress endpoints" that serve as "the unified entry point for the mail system," automatically handling load balancing and high availability (Amazon Web Services, 2025).
   - **AI-Powered Analysis:** It uses a machine learning model to scan the content, headers, links, and attachments of every email. Your description details a multi-faceted analysis:
     - Sender address spoofing detection (*"gcash-verify.net"* vs real GCash)
     - Link analysis and reputation checking (bit.ly masking)
     - Natural Language Processing for urgency and pressure tactics (*"URGENT"*, "suspended")
     - Threat intelligence feeds (checking against databases of known phishing links)
     - Behavioral pattern matching (*"Similar to attacks last week"*)

- **Action (Tiered Response):** It doesn't just block or pass instead it performs graduated actions:
    - **Marking:** Injects a visual warning (e.g.*"[SUSPICIOUS]"*) into the email's subject or body, which survives delivery to any device.
    - **Alerting:** Sends a real-time notification (e.g., text, dashboard alert) to the IT security team for high-confidence threats.
    - **Click-Time Protection:** If a user clicks a link in a previously analyzed email, the gateway can intercept that web request and block access to malicious sites in real-time.

3. **Architecture:**
    - It is a single system that can run on modest hardware. This centralization is intended to make the intervention economically and operationally feasible for a resource-constrained government agency.

Comparison

The current, decentralized, and device-dependent security posture (or lack thereof) of Philippine government agencies, which serves as the baseline for evaluating the effectiveness of the proposed AI gateway characterized by significant gaps:

1. **Do Nothing Approach:**
    - No technical controls are in place. Emails flow directly to the user's inbox, and success or failure depends entirely on the employee's ability to recognize a sophisticated phishing email.

    - This baseline scenario is alarmingly common across Philippine organizations. The PowerDMARC 2024 analysis of 1,650 domains across multiple sectors revealed that 78% of Philippine domains lack SPF records, 87.52% have no DMARC record at all, and less than 10% have correct DMARC configuration (PowerDMARC, 2024). For the government sector specifically, DMARC adoption is "very low," and 99.88% of Philippine domains have not implemented MTA-STS. Meaning most email transmissions occur without encryption or authentication protocols.

    - The consequences are stark. The Philippines experienced the highest number of cyberattacks in Southeast Asia in 2023, with 75% of organizations affected—significantly above the Asia-Pacific regional average of 59% (PowerDMARC, 2024). Without technical controls, organizations remain in a purely reactive posture where, as the National Privacy Commission's guidelines acknowledge, "personnel must immediately notify his or her immediate supervisor in case of a potential or actual personal data breach (NPC, 2021).

2. **Fragmented Tool Approach:**
    - An attempt to secure users by installing software on each endpoint. This includes browser extensions that check for malicious links, antivirus software with email scanning plugins, and operating system security features. However this approach is impossible to implement effectively in the context of the study; as it fails on old PCs. It also cannot be installed on personal phones, and creates an unmanageable maintenance burden.

- The limitations of this approach are well-documented in the Philippine context. The persistence of legacy systems means many government computers run Windows 7 or XP—operating systems that Microsoft no longer supports with security updates and that cannot run modern browser extensions or endpoint detection and response software. The Commission on Audit documented that 72.48% (P233.1 million) of the Office of the President's 2024 ICT budget remained unutilized, preventing necessary hardware upgrades (COA, 2025).

- For the BYOD population, legal barriers compound technical limitations. Under the Data Privacy Act (R.A. 10173), requiring employees to install device management software on personal phones creates "questionable" consent due to employer-employee power imbalances (Respicio & Co., 2024). Legal experts distinguish between defensible "light BYOD" or authenticator apps. Whilst legally risky "deep BYOD" or MDM with broad permissions cannot legally be enforced by the government.

- The maintenance burden is equally prohibitive. The DICT has only about 200 cybersecurity experts nationally, with an 80% overseas migration rate of Filipino talent, compared to Singapore's 2,000 experts (Uy, 2024). A 75% pay gap between government (P50,000) and private sector (P200,000) cybersecurity salaries makes retention impossible. Even if deployment were technically possible, the personnel required to manage fragmented endpoint security across hundreds of thousands of diverse devices simply does not exist.

3. **Outsourced/Cloud Approach:**
   - Purchasing a commercial email security solution from a large foreign vendor (e.g., proof-of-concept security from a major tech company). This is an alternative approach since the system proposed favors low-cost (no per-seat licenses). However it may lack context-awareness (trained on local Philippine scams, which foreign models may miss), and potentially be less explainable and customizable for local needs.

     The commercial alternative presents three distinct disadvantages in the Philippine government context:

- **Cost Barriers:** The economics are fundamentally misaligned. Commercial endpoint security licenses typically cost $50-150 per user annually. For a government with potentially hundreds of thousands of employees, this represents hundreds of millions of pesos in recurring costs which is unsustainable given the documented budget constraints where even critical agencies like the Anti-Money Laundering Council had their proposed P125.56 million cybersecurity budget cut to P93.01 million (Lacson, 2021).

- **Context Blindness:** Foreign-trained AI models may not recognize local phishing patterns. Philippine scams frequently target local brands (GCash, BDO, government agencies like BIR) using culturally-specific language and social engineering tactics. Generic models trained primarily on English-language, Western-brand phishing campaigns may have significantly higher false-negative rates for local threats. As PowerDMARC notes, the education sector shows only 1.39% correct DMARC implementation and healthcare is "the least performing" sector—areas where local threat actors concentrate their efforts (PowerDMARC, 2024).

- **Explainability and Control:** Commercial "black box" solutions often provide little visibility into why specific emails were flagged, making it difficult for government IT teams to validate decisions, tune models for local needs, or maintain audit trails required for public accountability. The proposed gateway's AI agent architecture achieves response times of 2-4 seconds while maintaining "explainable reasoning chains" (g0t-phish, 2025).

Output

The measurable improvements in security posture, operational efficiency, and user protection that result from deploying the centralized AI email gateway, compared to the status quo. They can be categorized into several key areas:

1. **Primary Security Outcomes**
    - **Increased Phishing Detection Rate:** The percentage of phishing emails successfully identified before reaching the user's inbox.
    - **Reduced Click-Through Rate on Malicious Links:** One goal is to prevent users from being harmed. By marking emails and providing click-time blocking, it aims to drastically reduce the number of users who fall for an attack.
    - **Decreased False Positive Rate:** Ideally the users or admins should not be overwhelmed with warnings on legitimate emails which the AI must be precised for.

2. **Operational & Administrative Outcomes**
    - **Universal Coverage Achieved:** The outcome of protecting 100% of users (old desktops, phones, tablets) with a single deployment. This is a unique and powerful outcome that no device-based solution can claim in this context. Directly addressing the population's heterogeneity documented by PIDS, which identified "lack of interoperability or lack of shared standards and compatible infrastructure across government agencies" as a major barrier (Serfino et al., 2020).
    - **Faster Incident Response Time:** Admins are alerted immediately upon a high-confidence threat, not hours or days later after a breach occurs. The dashboard provides a centralized view of all threats.
    - **Reduced Administrative Overhead:** The resource and manpower required to manage security is massively reduced. Instead of troubleshooting installations on multiple different devices. A security delegated administrator manages one server and responds to its alerts.

3. **User-Focused Outcomes**
    - **Seamless User Experience:** For 99% of email (the non-malicious kind), the user notices nothing different. The gateway is intended to be invisible.
    - **Just-in-Time User Training:** The visual warnings serve as a real-time teaching moment for employees, helping them become more security-conscious without formal training.

4. **Economic Outcomes**
    - **Zero Per-Device Cost:** The financial outcome is avoiding the huge capital and operational expenditure of endpoint security licenses. The cost is centralized and predictable with *"One server, One model"*. As Microsoft's architecture guidance confirms, centralized systems are driven by "business requirements associated with cost reduction" (Microsoft, 2005). The gateway's architecture is a single system on modest hardware. It is what Campbell and Solyanik (2002) describe as an "email concentrator" which makes it economically feasible where per-device licensing is not.

Context

The context of this study is the public sector digital communication environment of the Philippine government, where email remains one of the primary channels for official communication across agencies.

1. **Institutional Environment:**
    - **The Public Sector Landscape:** A massive network of national agencies (e.g., DICT, DepEd, DOST) and Local Government Units (LGUs) that rely on centralized government email services but operate under highly inconsistent security protocols. The Department of Information and Communications Technology itself has documented that government operations have historically been characterized by "siloed operations" and the absence of "centralized systems" (DICT, 2023). This institutional fragmentation creates exactly the "weakest link" problem where one poorly secured LGU account can become the entry point for attacks on national agencies.
    - **Infrastructure Heterogeneity:** A "digital divide" within the government itself, where some offices utilize modern cloud suites while others depend on legacy mail servers with outdated security patches. COA documented that the Office of the President left 72.48% (P233.1 million) of its 2024 ICT budget unutilized. Same funds that could have upgraded systems (COA, 2025). PowerDMARC revealed the security consequences: 78% of government domains lack SPF records, 87.52% have no DMARC record, and 99.88% lack MTA-STS encryption (PowerDMARC, 2024).
2. **Operational Constraints:**
    - **Fiscal Limitations:** Procurement is governed by strict public bidding laws and fixed annual budgets, making high-cost, per-user subscription models (SaaS) difficult to sustain for hundreds of thousands of employees. During Senate budget hearings, it was revealed that even critical agencies like the Anti-Money Laundering Council had their proposed P125.56 million cybersecurity budget cut to P93.01 million. A reduction of P32.55 million that the AMLC explicitly stated was needed for software to investigate cyber threats (Lacson, 2021). Senator Lacson noted these amounts are "smaller than the commissions and SOPs demanded by corrupt officials" in other agencies, highlighting how cybersecurity is systematically underfunded.

- **Policy Gaps:** A lack of enforceable "Minimum Security Standards" across all agencies, leading to a "weakest link" problem where one compromised LGU account can be used to spear-phish a national agency. The Cybercrime Investigation and Coordinating Center has emphasized that "technological advancements outpace current legislation" and that a "methodical" and "highly regulated" approach to technology governance is urgently needed (CICC, 2026).

3. **Threat Landscape:**
    - **Localized Phishing Campaigns:** Increasing attacks use local languages (Tagalog/Cebuano), regional themes (GCash, Landbank), and government-specific urgency ("Salary adjustment"). The NBI recently dismantled a major scamming operation in Pampanga targeting retirees with phishing links (NBI, 2026). The PNP-ACG recorded 29 cyber fraud incidents in early 2026 alone, with scammers using "social engineering" and stolen photos to pressure victims (PNP-ACG, 2026).
    - **Data Sovereignty and Sensitivity:** Protecting sensitive citizen data within a system that is currently a "soft target." DICT's Project SONAR discovered over 30,000 vulnerabilities across at least 2,002 government digital assets (DICT, 2024). The AMLC needs cybersecurity tools specifically "to assist our investigators in investigating virtual assets" linked to money laundering and terrorism (Lacson, 2021).
4. **Technological Readiness:**
    - **Shift Toward AI:** Growing institutional interest in automation to augment the severe lack of cybersecurity personnel. Only about 200 cybersecurity experts exist nationally, with an 80% overseas migration rate, compared to Singapore's 2,000 (Uy, 2024). A 75% pay gap between government (P50,000) and private sector (P200,000) makes retention impossible. The Government Cyber Resilience Summit 2026 even featured sessions on "AI-driven threat detection" and "harnessing AI-powered national defense" under the National Cybersecurity Plan (Manila Times, 2026). The CICC explicitly calls for moving the Philippines "from a reactive stance to a proactive one" through "methodical" AI employment (CICC, 2026).

Research Questions

RQ1: How does a centralized, server side AI intervention compare to decentralized, device dependent security scale in protecting different types of user groups (legacy, BYOD, and cross-platform)?

RQ2: What AI-driven techniques are currently employed in server-side email gateways to detect phishing and spoofing threats within resource-constrained public sector environments?

RQ3: In what ways do AI-powered visual warnings and click-time protection influence user security consciousness and behavior in the absence of formal training?

PICOC Keywords

The following keywords, derived directly from the research questions, are categorized according to the PICOC protocol to guide the systematic search for literature:

PICOC Element	Keywords from Research Questions
Population	"Resource-constrained public sector environments," "user groups," "legacy," "BYOD," "cross-platform"
Intervention	"Centralized server-side AI intervention," "server-side email gateways," "AI-powered visual warnings," "click-time protection"
Comparison	"Decentralized device-dependent security," "absence of formal training"
Outcome	"Protecting different types of user groups," "detect phishing and spoofing threats," "influence user security consciousness and behavior"
Context	"Public sector environments," "resource-constrained"

Inclusion and Exclusion Criteria

Inclusion Criteria

Studies included in this systematic literature review must meet the following conditions:

1. Studies published between 2021 and present to ensure up-to-date cybersecurity and AI developments.
2. Research focusing on AI, machine learning, or deep learning for email security.
3. Studies addressing phishing detection, spam filtering, or email threat detection.
4. Peer-reviewed journal articles, conference papers, or credible academic publications.
5. Research written in English.
6. Studies provide experimental results, framework, or evaluation metrics if applicable.
7. Studies that specifically discuss server-side or gateway-level interventions rather than just endpoint (PC/mobile) software.
8. Studies that evaluate detection of local or regionalized phishing scams, spoofing, or social engineering tactics.
9. Research discussing AI models that can be deployed at scale to protect a large, diverse population of users without needing access to individual devices.

Exclusive Criteria

Studies were excluded if they met any of the following conditions:

1. Research that is unrelated to email security or phishing detection.
2. Publications without empirical or research methodology.
3. Duplicate publications or preliminary versions of already published studies.
4. Exclude studies that focus on commercial tools requiring high per-user licensing fees.
5. Studies requiring specific modern hardware features (e.g., TPM chips or biometric sensors) that would exclude legacy users on Windows XP/7.
6. Research on AI models that are strictly trained and tested only on non-English datasets or specific foreign languages not relevant to Philippine communication.
7. Studies focusing on forensic analysis or "after-the-fact" detection that does not support your "Click-Time Protection" or real-time interception goals.
8. Research focusing solely on individual consumer-grade email (like personal Gmail/Outlook) rather than enterprise or institutional mail servers.
9. Research that requires administrative control or MDM (Mobile Device Management) enrollment for personal devices (BYOD).

Narrative of Literature Elimination

The elimination process was a critical step in refining the literature pool to ensure that the selected studies provide viable solutions for the Philippine public sector's unique constraints. The following justifications were used to exclude literature that, while technically sound, was practically inapplicable to this study

- **Exclusion of Device-Dependent/Endpoint Security:** Research focusing on antivirus software, browser extensions, or EDR (Endpoint Detection and Response) was eliminated. These solutions require administrative rights and modern OS compatibility, which are unavailable for the "Legacy" population using unpatchable Windows XP/7 systems and "BYOD" users where legal barriers prevent employer-mandated software installation.
- **Removal of Solutions Requiring Administrative Enrollment (MDM):** Any literature requiring Mobile Device Management (MDM) for personal devices was excluded. This is based on the legal reality in the Philippines where "deep BYOD" creates questionable consent under the Data Privacy Act (R.A. 10173), making such interventions legally unfeasible for government agencies.
- **Elimination of Forensic or "After-the-Fact" Detection:** Studies that focused on post-breach analysis were removed in favor of real-time interception. To address **RQ3**, the review prioritized "Click-Time Protection" and "Visual Warnings" that occur before a user interacts with a threat.
- **Filtering of High-Cost/Per-User Subscription Models:** Literature describing commercial frameworks with "per-seat" licensing was eliminated. This exclusion is justified by the documented budget constraints in the Philippine government, where even critical security budgets are frequently cut, making a "Zero Per-Device Cost" centralized gateway the only economically sustainable option.
- **Context-Agnostic Models:** Research on AI models trained strictly on Western or non-English datasets was deprioritized. This ensures the selected literature supports the detection of localized phishing patterns targeting Philippine brands like GCash or local government memoranda.

Quality Assurance (QA) Checklist

Ensuring the reliability and validity of the selected research studies, a Quality Assessment (QA) checklist was applied:

Scoring Rubrics:

Level of Participation	
1	Not, and not considered (Score:0)
2	Partially (Score:0.5)
3	Yes (Score:1)

Level of Agreement	
1	Disagree (Score:1)
2	Somewhat disagree (Score:2)
3	Neither agree nor disagree (Score:3)
4	Somewhat agree (Score:4)
5	Agree (Score:5)

QA Criteria Evaluation Table:

ID	Quality Assessment Question	Level of Participation	Level of Agreement
QA1	Does the study explicitly define the AI or Machine Learning architecture used for email security?		
QA2	Does the study provide a clear experimental setup or reproducible evaluation method?		

QA3	Are the datasets or testing environments (e.g., real-world vs. synthetic) clearly described?		
QA4	Is the research applicable to organizational or public-sector email infrastructures?		
QA5	Does the study address the scalability of the model for high-volume institutional traffic?		
QA6	Are performance metrics (Accuracy, Precision, Recall, and False Positive Rate) clearly reported?		
QA7	Does the research provide details on real-time interception or "Click-Time" protection?		
QA8	Is the model's "explainability" or the method for visual user-warning discussed?		
QA9	Does the study discuss limitations or threats to validity (reliability) of the results?		
Total	**Weighted Quality Score**	**/ 9.0**	**/ 45.0**

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